LG LENSGUARD

HOW IT WORKS

A phone website that helps you look for hidden cameras.

You open LensGuard on your phone, point the camera around the room, and read a short result. The result tells you what is worth a closer look. It does not tell you the room is safe, and it does not tell you a spot is definitely a camera.

Anyone can use it. You do not create an account. You do not install an app. You open the site, allow the camera, and start a check.

The video stays on your phone. After the check, the phone can send a short written summary to be saved. The summary is words and numbers, such as "optical check, 2 spots, medium confidence." It is not the video.

What you get

A calm second look. Bright reflections, odd lights, and unknown names on a Wi-Fi list are clues. You still decide what they mean by looking at the real object.

The four parts

LensGuard is four pieces working together. You only touch the first one.

1 · WEBSITE

The screens you tap

The website is built with **Next.js**. That is a tool for making websites. On your phone it is just a site in the **browser** – the app you already use to open the web, such as Safari or Chrome.

2 · PHONE

The camera check

The check runs on the phone. The camera turns on, the phone studies what it sees, and the video never leaves the device. A bright spot on the screen is marked so you can walk over and look at it.

3 · SERVER

The computer that receives the summary

A **server** is a computer on the internet that waits for messages. LensGuard's server is built with **Node.js** and **Express**. It accepts the written summary. It does not watch the camera.

4 · DATABASE

The place the summary is kept

A **database** is an organized store of records. LensGuard uses **PostgreSQL**, a **relational** database. Relational means records can point to each other, the way "this summary belongs to this visit" points from one record to another.

FLOW

What you do, from open to result

A check is one pass around the room. The system calls each check a scan. You can do more than one during the same visit.

1. **Open the site**

   You go to LensGuard in the phone browser. There is no sign-up screen.

2. **Tap Start**

   This begins a room inspection. The site asks the browser for the camera.

3. **Allow the camera**

   The phone shows its usual permission question. If you say no, the check cannot start. If you say yes, you see a live picture from the rear camera.

4. **Pick a check**

   Optical looks for lens-like reflections. Infrared looks for a steady bright dot in low light. Network walks you through the list of devices on the Wi-Fi router. You can run one, or all three.

5. **Move through the room**

   You pan the phone slowly. The picture stays on the phone while the check runs.

6. **Read the result**

   The screen lists clues, such as “2 possible reflections” or “no infrared indicator.” The wording asks you to review them. It does not declare a camera found, and it does not declare the room clear.

7. **Keep the summary, or delete it**

   The written summary can be saved so you can open it again on this phone. You can also delete it. Deleting removes the saved summary. It does not change what you already saw in the room.

ARCHITECTURE

Where each thing lives

The phone does the looking. The server stores a note about the looking. The database files that note so it can be read again.

PHONE · YOUR BROWSER

Website and camera check

You see the LensGuard screens here. The camera opens here. The video is studied here. Nothing about the picture is uploaded while you scan.

↓ written summary only ↓

SERVER · NODE.JS AND EXPRESS

Receives the summary

The summary includes the kind of check, how long it took, how many spots were flagged, and a confidence word: low, medium, or high. Confidence means how strong the clue looks. It is not proof.

↓ one saved record ↓

DATABASE · POSTGRESQL

Keeps the summary

Each finished check becomes one record. A record is a single saved row of facts. You can open that record later, or delete it.

> The video, the live camera picture, and the frame-by-frame analysis stay inside the phone box. They are not copied into the server or the database.

ONE CHECK

From the first tap to a saved note

1. **Start**

   You tap Start and choose a check. The server opens a new scan and returns an id, which is a unique name for this check.

2. **Look, on the phone**

   The camera runs. Spots are marked on your screen. The server is idle during this part because the picture never goes to it.

3. **Send the summary**

   When you finish, the phone sends the written result. The server attaches it to the scan that was opened in step 1.

4. **Read it again**

   You can ask for that same scan. The server reads the record and shows the summary.

5. **Delete it**

   You can remove the record. After that, the summary is gone from the database.

What the server does at each step

- **Create scan** when you start.
- **Save result** when the check finishes.
- **Read scan** when you open a summary again.
- **Delete scan** when you remove it.

A separate health check only answers “the server is on.” It does not store anything about your room.

THE THREE CHECKS

Three ways to look, three different clues

Each check answers a different question. None of them, alone or together, proves that a camera is present or absent.

OPTICAL

A small bright reflection that might be a lens

You dim the room, turn on the phone light if the browser allows it, and move the camera slowly across smoke detectors, clocks, vents, and mirror frames. A camera lens can throw back a small, tight glint. Screws, glass, and polished plastic can do the same. LensGuard marks the glint and asks you to inspect that object yourself.

INFRARED

A steady bright dot in a dark room

Some cameras use infrared light, which your eyes cannot see, so they can record in the dark. Some phone cameras can still pick that light up as a bright point. You darken the room and look for a dot that stays put. Many phones filter this light out, so a clear screen does not mean the room has no infrared camera. It means this phone did not see one.

NETWORK

A guided look at the Wi-Fi device list

A website on a phone cannot reliably search the whole network by itself. LensGuard instead walks you through your router's list of connected devices. You read the names. An unknown name is a reason to ask what the device is. It is not, by itself, a hidden camera. You can type a device name or address into the checklist if you want that note saved with the summary.

USE CASES

When a person would use it

HOTEL OR RENTAL, BEFORE UNPACKING

Sam wants a first look around the room

Sam opens LensGuard, allows the camera, and chooses the optical check. Sam pans across the alarm clock, the TV, and the smoke detector. Two reflections are marked. Sam walks up to both objects and looks at them. The video stays on the phone. The saved summary says: optical check, 2 spots, and a confidence word.

WHAT SAM TAPS
Start, then Optical

WHAT STAYS ON THE PHONE
The camera video

WHAT CAN BE SAVED
The written summary

WHAT SAM STILL DOES
Looks at the real objects

DARK ROOM

Sam turns the lights off and tries the infrared check

Sam chooses the infrared check and moves the phone along the ceiling line. No steady bright dot appears. The result says no infrared indicator was seen on this phone. Sam can still do the optical check, because the two checks look for different clues.

WHAT SAM TAPS
Infrared

WHAT STAYS ON THE PHONE
The dark-room video

WHAT CAN BE SAVED
“No indicator detected”

WHAT IT DOES NOT MEAN
The room is certified clear

WI-FI LIST

Sam checks the router with the network guide

Sam opens the router page, follows the LensGuard checklist, and sees a device named only with a string of letters. Sam notes the name in the guide. LensGuard does not connect to the router and does not label that device a camera. The note can be saved with the other summaries from this visit.

WHAT SAM TAPS
Network inspection

WHAT STAYS ON THE PHONE
No camera video is taken

WHAT CAN BE SAVED
The checklist note

WHAT SAM STILL DOES
Asks what the device is

SAME PHONE, LATER

Sam opens the summaries from this visit

The browser kept a session id, a random code that groups this visit's checks without an account. Sam opens the list and sees the optical summary, the infrared summary, and the network note. Another person's phone does not show Sam's list, because it has a different session id.

WHAT SAM TAPS
Past summaries

WHAT STAYS ON THE PHONE
The session id in the browser

WHAT IS READ
Saved summaries for this visit

ACCOUNT NEEDED
No

REMOVING A SUMMARY

Sam deletes the saved note

Sam opens one summary and taps delete. That record is removed from the database. The other checks from the visit remain until Sam deletes those too. The room itself is unchanged. Deleting a summary only deletes the note.

WHAT SAM TAPS
Delete on one summary

WHAT STAYS ON THE PHONE
Nothing new is recorded

WHAT IS REMOVED
That one database record

WHAT REMAINS
Any summary Sam did not delete

WHAT GETS SAVED

One visit can hold many checks

The database does not store a picture of the room. It stores one record per check. The records from a single visit share one session id, so they can be listed together.

One visit on this phone

> Optical check — 2 possible reflections — medium confidence
>
> Infrared check — no indicator detected
>
> Network check — one unknown device name noted

Each of those lines is stored as its own record, with the same fields:

Id	A unique name for this check, created when it starts.
Session id	A random code in the browser that groups checks from one visit. It is not your name, email, or account.
Check type	Optical, infrared, or network.
Start time	When the check began.
Duration	How long the check took.
Confidence	Low, medium, or high. This describes the clue, not a final verdict.
Result note	A short written result, such as how many spots were marked or that the network list was reviewed.

A fuller written report, and an optional account for people who later want their history on a second phone, can be added beside these records. The flow in this guide works without either of them.

PRIVACY

Three rules the product follows

RULE 1

The video stays on the phone

Live pictures are processed in the browser. They are not uploaded for the check to work.

RULE 2

The server stores the summary

What leaves the phone is a short note: type, time, count, confidence, and a few words about the result.

RULE 3

No account is required

You can start a check immediately. The session id is only there so this phone can find its own summaries.

Words used in this guide

Browser	The app that opens websites. LensGuard runs inside it.
Check / scan	One pass of one kind: optical, infrared, or network. “Scan” is the name stored in the system.
Summary	The written result. It is what can be saved. The video is not part of it.
Server	The internet computer that receives a summary, stores it, reads it back, or deletes it.
Database	The organized store of those summaries. PostgreSQL is the database LensGuard uses.
Record	One saved summary. One check produces one record.
Session id	A random code that ties this phone’s checks together without asking who you are.
Relational	A database style where records can point to each other. One visit points to many checks.
Confidence	How strong a clue looks: low, medium, or high. It is not a yes-or-no answer.